

Anurag Krishna, Ph.D.

Photovoltaics Material and Device Expert

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DATE OF BIRTH: 03/03/1991 | **NATIONALITY:** INDIAN

SUMMARY

Thin-film photovoltaic device engineer with over 12 years of international experience in designing, fabricating, and characterizing large-area semiconductor stacks for high-efficiency solar modules. Expertise in perovskite and organic materials, interface engineering, and long-term reliability from cell to module scale. Proven success in leading European and industrial R&D programs, defining performance and stability standards, and transferring lab-scale innovations to scalable manufacturing in partnership with industry and equipment suppliers. Strong background in data-driven analysis of device degradation, outdoor performance monitoring, and machine-learning-based performance modeling.

SKILLS

- **Photovoltaic devices:** Design and fabrication of perovskite and organic solar cells; interface, transport-layer, and encapsulation optimization for efficiency and stability.
- **Characterization:** J–V/IV, EQE/IPCE, PL/TRPL, EL, impedance, transient measurements, and device/module performance mapping.
- **Reliability:** Accelerated light, heat, and bias testing; IEC/ISOS protocols; outdoor monitoring and degradation analysis.
- **Processing & scale-up:** Solution and vacuum deposition, film morphology control, module design, interconnection, and yield optimization.
- **Data & AI for materials:** Large-scale structure–property analysis; LLM-based literature and data extraction; automated degradation analysis and ML-based lifetime prediction.
- **Funding & project management:** Horizon Europe, FWO, and industrial projects; coordination of research consortia and industry collaborations.
- **Leadership & communication:** Supervision of researchers and engineers; university teaching, mentoring, scientific writing, public speaking, and interdisciplinary collaboration.

WORK EXPERIENCE

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| 01/2022-Present | Interuniversity Microelectronics Centre (IMEC), Belgium
R&D Project Leader, Thin Film PV <ul style="list-style-type: none">• R&D roadmapping, module upscaling, and reliability of perovskite photovoltaics.• Lead European, national, and industrial projects with successful funding acquisition (Horizon Europe, EIC, FWO, industry B2B/licensing).• Drive spin-off and venture initiative, preparing investor pitches and pathways for pilot manufacturing.• Supervise engineers, PhD students, and postdocs, fostering technical expertise and career development.• Active in policy and outreach, contributing to ETIP PV roadmaps, advising CINEA, and founding workshops bridging science, policy, and industry. |
| 09/2019-09/2021 | École Polytechnique Fédérale de Lausanne (EPFL), Switzerland
Scientist (Marie-Curie Fellow)
Advisors: Prof. Michael Grätzel and Prof. Anders Hagfeldt <ul style="list-style-type: none">• Awarded the Marie Skłodowska-Curie Fellowship (EU) for perovskite photovoltaics research.• Developed nanoscale interface engineering achieving ~25% efficiency with best-in-class stability.• Advanced understanding of defect passivation, ion migration, and stability mechanisms in perovskites.• Published 15+ papers in leading journals and delivered talks at major conferences (E-MRS, EU PVSEC, HOPV).• Managed the stability lab, mentored PhD/Master's students, and contributed to successful EU/Swiss grants. |

04/2017-07/2019

Centre National de la Recherche Scientifique (CNRS)

CNRS Postdoctoral Fellow

Advisor: Dr. Frédéric Sauvage

- Launched perovskite PV research at CNRS-LRCS, setting up new lab facilities and achieving benchmark device efficiencies.
- Developed 3D/low-dimensional heterostructures to enhance stability and led in-situ studies (XRD, TEM, NMR) on degradation mechanisms.
- Supervised a PhD student and gained strong expertise in crystallography, microscopy, and ultrafast photophysics.

EDUCATION

08/2012–02/2017

Nanyang Technological University, Singapore

Ph.D. in Materials Science & Engineering

Advisors: Dr. Andrew C. Grimsdale and Dr. Cesare Soci

Thesis: Hole Transporting Materials for $\text{CH}_3\text{NH}_3\text{PbX}_3$ Perovskite Solar Cells

08/2008–06/2012

Delhi College of Engineering (now Delhi Technological University), India

Bachelor of Engineering in Polymer Science & Chemical Technology (First class)

Thesis on the synthesis and characterization of conducting polymers nanostructures at the University of Auckland, New Zealand

PUBLICATIONS

1. Y. Xu, Z. Wang, C. Deng, L. Qi, Y. Mo, Q. Wang, D. Yu, Y. Wang, Y. Chen, Y. Tian, P. Zhao, M. Luo, D. Jiang, H. Li, X. Ding, R. Xia, D. Yan, P. Ge, D. Xin, H. Yao, S. Zhu, K. Xian, C. Zhou, G. Yang, L. Yin, Y. Yang, **Anurag Krishna**, W. C. H. Choy, P. P. Altermatt, J. Wang, Z. Xie, X. Zhang, J. Chu, H. Zhang, J. Gao, Y. Chen, "Defect passivation and optical management of triple-junction solar cells" **Nature**, 2026. <https://doi.org/10.1038/s41586-026-11010-8>
2. Amit Kumar Harit, Zi-Fan He, Yinghuan Kaung, Tamara Merckx, Aranzazu Aguirre, Buyi Yan, Yi Hou, Hairan Tan, Tom Aernouts, **Anurag Krishna**, "Taking Perovskite Photovoltaics From Promise to Product" **Nature Review Clean Technology**, 2, 453–466 (2026). doi.org/10.1038/s44359-026-00173-2
3. Jonathan Parion, Baptiste Jacquet, Dawar Ali, Raju Pusapati, Amit Kumar Harit, Aurora Rizzo, Yinghuan Kuang, Filip Duerinckx, Hariharsudan Sivaramakrishnan Radhakrishnan, Tom Aernouts, **Anurag Krishna**, Jef Poortmans, Johan Lauwaert, Bart Vermang, "In-depth study of degradation in scalable wide bandgap perovskite cells" **Mater. Futures**, 2025, 4 04510. [Impact factor = 10.8] <https://iopscience.iop.org/collections/mf-250108-752>
4. Jeremy Hieulle, **Anurag Krishna**, Hazem Adel Musallam, Tom Aernouts, Alex Redinger, "Modeling the FA and I losses in mixed-halide perovskite through chemical rate equations: insights into light-induced degradation", **EES Solar.**, 2025, 1 (4) 645–658. <https://doi.org/10.1039/d5el00058k>
5. Tzu-Sen Su, **Anurag Krishna**, Chenxu Zhao, Junhao Chu, Hong Zhang, "Supramolecular engineering in hybrid perovskite optoelectronics", **Chem. Soc. Rev.**, 2025, 54 (13) 6448–6481. [Impact factor = 39] <https://doi.org/10.1039/d4cs00924j>
6. Vasiliki Paraskeva, Matthew Norton, Andreas Livera, Andreas Kyprianou, Maria Hadjipanayi, Elias Peraticos, Aranzazu Aguirre, Santhosh Ramesh, Tamara Merckx, Rita Ebner, Tom Aernouts, **Anurag Krishna**, George E. Georgiou, "Diurnal Changes and Machine Learning Analysis of Perovskite Modules Based on Two Years of Outdoor Monitoring", **ACS Energy Lett.**, 2024, 9 (10) 5081–5091. [Impact factor = 18.2] <https://doi.org/10.1021/acsenerylett.4c01943>
7. Oleksandr Volochanskyi, Golam Haider, Essa A. Alharbi, George Kakavelakis, Martin Mergl, Mukesh Kumar Thakur, **Anurag Krishna**, Michael Graetzel, Martin Kalbáč, "Graphene-Templated Achiral Hybrid Perovskite for Circularly Polarized Light Sensing", **ACS Appl. Mater. Interfaces.**, 2024, 16 (39) 52789–52798. [Impact factor = 8.2] <https://doi.org/10.1021/acsaami.4c10289>
8. Rashmi Runjhun, Essa A. Alharbi, Zygmunt Drużyński, **Anurag Krishna**, Małgorzata Wolska-Pietkiewicz, Viktor Škorjanc, Thomas P. Baumeler, George Kakavelakis, Felix Eickemeyer, Mounir Mensi, Shaik M. Zakeeruddin, Michael Graetzel, Janusz Lewiński, "High-Performance Perovskite Solar Cells with Zwitterion-Capped-ZnO Quantum Dots as Electron Transport Layer and NH_4X (X = F, Cl, Br) Assisted Interfacial Engineering", **Energy Environ. Mater.**, 2024, 7 (5). [Impact factor = 13] <https://doi.org/10.1002/eeem2.12720>
9. Xinxin Lian, Ye Xu, Wei Fu, Rui Meng, Quanxing Ma, Chunyu Xu, Ming Luo, Ying Hu, Junchao Han, Hao Min, **Anurag Krishna**, Yifan Chen, Huawei Zhou, Xueling Zhang, Cong Chen, Jin Chang, Can Li, Yifeng Chen, Zhiqiang Feng,

- Zhen Li, Guangzheng Zuo, Jifan Gao, Hong Zhang, Xiaoliang Mo, Junhao Chu, "Homogenizing Morphology and Composition of Methylammonium-Free Wide-Bandgap Perovskite for Efficient and Stable Tandem Solar Cells", **Adv. Funct. Mater.**, 2024, 34 (37). [Impact factor = 19] <https://doi.org/10.1002/adfm.202402061>
10. Stijn Lenaers, Stijn Lammar, **Anurag Krishna**, Valerio Stacchini, Tom Cardeynals, Huguette Penxten, Christ Weijtens, Michael Verhage, Bart Ruttens, Wouter Maes, Jan D'Haen, Artem Musiienko, Tom Aernouts, Laurence Lutsen, Dirk Vanderzande, Jef Poortmans, Wouter Van Gompel, "Pyrene-Based Self-Assembled Monolayer with Improved Surface Coverage and Energy Level Alignment for Perovskite Solar Cells", **Adv. Funct. Mater.**, 2024. [Impact factor = 19] <https://doi.org/10.1002/adfm.202411922>
 11. Himanshu Phirke, S. Gharabeiki, Ajay Singh, **Anurag Krishna**, Sussane Siebentritt, Alex Redinger, "Quantifying recombination and charge carrier extraction in halide perovskites via hyperspectral time-resolved photoluminescence imaging", **APL Energy.**, 2024, 2 (1). [Impact factor = 3.6] <https://doi.org/10.1063/5.0188166>
 12. Chenxu Zhao, Qi Zhang, Yongao Lyu, Jing Liu, Fan Shen, Huijing Liu, Hao Kong, Huifang Han, **Anurag Krishna**, Jia Xu, Hong Zhang, Jianxi Yao, "Regulating the Crystallization of FAPbI₃-Based Perovskite with a Furan Substituted Ethylammonium Additive for Achieving Highly Efficient Solar Cells", **Adv. Funct. Mater.**, 2024, 34 (41). [Impact factor = 19] <https://doi.org/10.1002/adfm.202404099>
 13. Chenxu Zhao, Zhiwen Zhou, Masaud Almalki, Michael A. Hope, Jiashang Zhao, Thibaut Gallet, **Anurag Krishna**, Aditya Mishra, Felix T. Eickemeyer, Jia Xu, Yingguo Yang, Shaik M. Zakeeruddin, Alex Redinger, Tom J. Savenije, Lyndon Emsley, Jianxi Yao, Hong Zhang, Michael Grätzel, "Stabilization of highly efficient perovskite solar cells with a tailored supramolecular interface", **Nat. Commun.**, 2024, 15 (1). [Impact factor = 15.7] <https://doi.org/10.1038/s41467-024-51550-z>
 14. Jeremy Hieulle, **Anurag Krishna**, Ariadni Boziki, Jean-Nicolas Audinot, Muhammad Uzair Farooq, Joana Ferreira Machado, Marko Mladenović, Himanshu Phirke, Ajay Singh, Tom Wirtz, Alexandre Tkatchenko, Michael Graetzel, Anders Hagfeldt, Alex Redinger, "Understanding and decoupling the role of wavelength and defects in light-induced degradation of metal-halide perovskites", **Energy Environ. Sci.**, 2024, 17 (1) 284–295. [Impact factor = 30.8] <https://doi.org/10.1039/d3ee03511e>
 15. Chenxu Zhao, Hong Zhang, **Anurag Krishna**, Jia Xu, Jianxi Yao, "Interface Engineering for Highly Efficient and Stable Perovskite Solar Cells", **Adv. Opt. Mater.**, 2023, 12 (7). [Impact factor = 7.2] <https://doi.org/10.1002/adom.202301949>
 16. Xin Zhang, Weiming Qiu, Sofia Apergi, Shivam Singh, Paulo Marchezi, Wenya Song, Christian Sternemann, Karim Elkhoully, Dong Zhang, Aranzazu Aguirre, Tamara Merckx, **Anurag Krishna**, Yuanyuan Shi, Andrea Bracesco, Cristian van Helvoirt, Frennie Bens, Valerio Zardetto, Jan D'Haen, Anran Yu, Geert Brocks, Tom Aernouts, Ellen Moons, Shuxia Tao, Yiqiang Zhan, Yinghuan Kuang, Jef Poortmans, "Minimizing the Interface-Driven Losses in Inverted Perovskite Solar Cells and Modules", **ACS Energy Lett.**, 2023, 8 (6) 2532–2542. [Impact factor = 18.2] <https://doi.org/10.1021/acsenenergylett.3c00697>
 17. **Anurag Krishna**, Viktor Škorjanc, Mathias Dankl, Jeremy Hieulle, Himanshu Phirke, Ajay Singh, Essa A. Alharbi, Hong Zhang, Felix T. Eickemeyer, Shaik M. Zakeeruddin, G. N. Manjunatha Reddy, Alex Redinger, Michael Grätzel, Anders Hagfeldt, "Mitigating the Heterointerface Driven Instability in Perovskite Photovoltaics", **ACS Energy Lett.**, 2023, 8 (8) 3604–3613. [Impact factor = 18.2] <https://doi.org/10.1021/acsenenergylett.3c01029>
 18. Stijn Lammar, Wouter T. M. Van Gompel, Stijn Lenaers, Martijn Mertens, Hans-Gerd Boyen, Derese Desta, Afshin Hadipour, Laurence Lutsen, Dirk Vanderzande, **Anurag Krishna**, Yaser Abdulraheem, Tom Aernouts, Jeff Poortmans, "Organic ammonium iodide salts as passivation for buried interface enables efficient and stable NiO_x-based p-i-n perovskite solar cells", **J. Mater. Chem. C**, 2023. [Impact factor = 5.1] <https://doi.org/10.1039/d2tc04215k>
 19. Chenxu Zhao, Hong Zhang, Masaud Almalki, Jia Xu, **Anurag Krishna**, Felix T. Eickemeyer, Jing Gao, Yu Mao Wu, Shaik M. Zakeeruddin, Junhao Chu, Jianxi Yao, Michael Grätzel, "Stabilization of FAPbI₃ with multifunctional alkali-functionalized polymer", **Adv. Mater.**, 2023. [Impact factor = 26.8] <https://doi.org/10.1002/adma.202211619>
 20. Tamara Merckx, Aranzazu Aguirre, Yinghuan Kuang, Arvid van der Heide, Ali Hajjiah, Yaser Abdulraheem, **Anurag Krishna**, Jef Poortmans, Tom Aernouts, "Stable Device Architecture With Industrially Scalable Processes for Realizing Efficient 784 cm² Monolithic Perovskite Solar Modules", **IEEE J. Photovolt.**, 2023, 13 (3) 419–421. [Impact factor = 2.6] <https://doi.org/10.1109/jphotov.2023.3259061>
 21. Xin Zhang, Stijn Eurelings, Andrea Bracesco, Wenya Song, Stijn Lenaers, Wouter Van Gompel, **Anurag Krishna**, Tom Aernouts, Laurence Lutsen, Dirk Vanderzande, Mariadriana Creatore, Yiqiang Zhan, Yinghuan Kuang, Jef Poortmans, "Surface Modulation via Conjugated Bithiophene Ammonium Salt for Efficient Inverted Perovskite Solar

- Cells”, **ACS Appl. Mater. Interfaces.**, 2023, 15 (40) 46803–46811. [Impact factor = 8.2] <https://doi.org/10.1021/acsaami.3c08119>
22. Hong Zhang, Kasra Darabi, Narges Yaghoobi Nia, **Anurag Krishna**, Paramvir Ahlawat, Boyu Guo, Masaud Hassan S. Almalki, Tzu-Sen Su, Dan Ren, Viacheslav Bolnykh, Luigi Angelo Castriotta, Mahmoud Zendehele, Lingfeng Pan, Sandy Sanchez Alonso, Ruipeng Li, Shaik M. Zakeeruddin, Anders Hagfeldt, Ursula Rothlisberger, Aldo Di Carlo, Aram Amassian, Michael Grätzel, “A universal co-solvent dilution strategy enables facile and cost-effective fabrication of perovskite photovoltaics”, **Nat. Commun.**, 2022, 13 (1). [Impact factor = 15.7] <https://doi.org/10.1038/s41467-021-27740-4>
 23. Essa A. Alharbi, **Anurag Krishna**, Nikolaos Lempesis, Mathias Dankl, Irea Mosquera-Lois, Michael A. Hope, Thomas P. Baumeler, George Kakavelakis, Aditya Mishra, Felix T. Eickemeyer, Olivier Ouellette, Thanyarat Chawanpunyawat, Anders Hagfeldt, Shaik M. Zakeeruddin, Lyndon Emsley, Lukas Pfeifer, Ursula Rothlisberger, Michael Grätzel, “Cooperative passivation of perovskite solar cells by alkyltrimethylammonium halide amphiphiles”, **Joule.**, 2022, 7 (1), 183-200. [Impact factor = 35.4] <https://doi.org/10.1016/j.joule.2022.11.013>
 24. Essa A. Alharbi, Thomas P. Baumeler, **Anurag Krishna**, Ahmed Y. Alyamani, Felix T. Eickemeyer, Olivier Ouellette, Linfeng Pan, Fahad S. Alghamdi, Zaiwei Wang, Mohammad Hayal Alotaibi, Bowen Yang, Masaud Almalki, Mounir D. Mensi, Hamad Albrithen, Abdulrahman Albadri, Anders Hagfeldt, Shaik M. Zakeeruddin, Michael Grätzel, “Formation of High-Performance Multi-Cation Halide Perovskites Photovoltaics by δ -CsPbI₃/ δ -RbPbI₃ Seed-Assisted Heterogeneous Nucleation”, **Adv. Energy Mater.**, 2021, 11 (16) 2003785. [Impact factor = 26] <https://doi.org/10.1002/aenm.202003785>
 25. Essa A. Alharbi, **Anurag Krishna**, Thomas P. Baumeler, Mathias Dankl, George C. Fish, Felix Eickemeyer, Olivier Ouellette, Paramvir Ahlawat, Viktor Škorjanc, Elsa John, Bowen Yang, Lukas Pfeifer, Claudia Esther Avalos, Linfeng Pan, Mounir Mensi, Pascal Alexander Schouwink, Jacques-E. Moser, Anders Hagfeldt, Ursula Rothlisberger, Shaik M. Zakeeruddin, Michael Grätzel, “Methylammonium Triiodide for Defect Engineering of High-Efficiency Perovskite Solar Cells”, **ACS Energy Lett.**, 2021, 6 (10) 3650–3660. [Impact factor = 18.2] <https://doi.org/10.1021/acsaenergylett.1c01754>
 26. Hong Zhang, Felix Thomas Eickemeyer, Zhiwen Zhou, Marko Mladenović, Farzaneh Jahanbakhshi, Lena Merten, Alexander Hinderhofer, Michael A. Hope, Olivier Ouellette, Aditya Mishra, Paramvir Ahlawat, Dan Ren, Tzu-Sen Su, **Anurag Krishna**, Zaiwei Wang, Zhaowen Dong, Jinming Guo, Shaik M. Zakeeruddin, Frank Schreiber, Anders Hagfeldt, Lyndon Emsley, Ursula Rothlisberger, Jovana V. Milić, Michael Grätzel, “Multimodal host–guest complexation for efficient and stable perovskite photovoltaics”, **Nat. Commun.**, 2021, 12 (1). [Impact factor = 15.7] <https://doi.org/10.1038/s41467-021-23566-2>
 27. **Anurag Krishna**, Hong Zhang, Zhiwen Zhou, Thibaut Gallet, Mathias Dankl, Olivier Ouellette, Felix T. Eickemeyer, Fan Fu, Sandy Sanchez, Mounir Mensi, Shaik M. Zakeeruddin, Ursula Rothlisberger, G. N. Manjunatha Reddy, Alex Redinger, Michael Grätzel, Anders Hagfeldt, “Nanoscale interfacial engineering enables highly stable and efficient perovskite photovoltaics”, **Energy Environ. Sci.**, 2021, 14 (10) 5552–5562. [Impact factor = 30.8] <https://doi.org/10.1039/d1ee02454j>
 28. Xufeng Ling, Hongwei Zhu, Weidong Xu, Cheng Liu, Linfeng Pan, Dan Ren, Jianyu Yuan, Bryon W. Larson, Carole Grätzel, Ahmad R. Kirmani, Olivier Ouellette, **Anurag Krishna**, Jianguo Sun, Chunyang Zhang, Youyong Li, Shaik M. Zakeeruddin, Jing Gao, Yuhang Liu, James R. Durrant, Joseph M. Luther, Wanli Ma, Michael Grätzel, “Combined Precursor Engineering and Grain Anchoring Leading to MA-Free”, **Angew. Chem. Int. Ed.**, 2021, 133 (52) 27505–27512. [Impact factor = 16.9] <https://doi.org/10.1002/ange.202112555>
 29. Haizhou Lu, **Anurag Krishna**, Shaik M. Zakeeruddin, Michael Grätzel, Anders Hagfeldt, “Compositional and Interface Engineering of Organic-Inorganic Lead Halide Perovskite Solar Cells”, **iScience.**, 2020, 23 (8) 101359. [Impact factor = 4.1] <https://doi.org/10.1016/j.isci.2020.101359>
 30. Tzu-Sen Su, Felix Thomas Eickemeyer, Michael A. Hope, Farzaneh Jahanbakhshi, Marko Mladenović, Jun Li, Zhiwen Zhou, Aditya Mishra, Jun-Ho Yum, Dan Ren, **Anurag Krishna**, Olivier Ouellette, Tzu-Chien Wei, Hua Zhou, Hsin-Hsiang Huang, Mounir Driss Mensi, Kevin Sivula, Shaik M. Zakeeruddin, Jovana V. Milić, Anders Hagfeldt, Ursula Rothlisberger, Lyndon Emsley, Hong Zhang, Michael Grätzel, “Crown Ether Modulation Enables over 23% Efficient Formamidinium-Based Perovskite Solar Cells”, **J. Am. Chem. Soc.**, 2020, 142, 47, 19980–19991 [Impact factor = 15.6] <https://doi.org/10.1021/jacs.0c08592>
 31. **Anurag Krishna**, Mohammad Ali Akhavan Kazemi, Michel Sliwa, G. N. Manjunatha Reddy, Laurent Delevoye, Olivier Lafon, Alexandre Felten, Mai Trang Do, Sébastien Gottis, Frédéric Sauvage, “Defect Passivation via the Incorporation of Tetrapropylammonium Cation Leading to Stability Enhancement in Lead Halide Perovskite”, **Adv. Funct. Mater.**, 2020, 30 (13) 1909737. [Impact factor = 19] <https://doi.org/10.1002/adfm.201909737>

32. Mohammad Ali Akhavan Kazemi, Parth Raval, Kirill Cherednichekno, Jean-Noel Chotard, **Anurag Krishna**, Arnaud Demortiere, G. N. Manjunatha Reddy, Frédéric Sauvage, "Molecular-Level Insight into Correlation between Surface Defects and Stability of Methylammonium Lead Halide Perovskite Under Controlled Humidity", **Small Methods.**, 2020, 5 (2) 2000834. [Impact factor = 9.1] <https://doi.org/10.1002/smt.202000834>
33. **Anurag Krishna**, Sébastien Gottis, Mohammad Khaja Nazeeruddin, Frédéric Sauvage, "Mixed Dimensional 2D/3D Hybrid Perovskite Absorbers: The Future of Perovskite Solar Cells?", **Adv. Funct. Mater.**, 2018, 29 (8). [Impact factor = 19] <https://doi.org/10.1002/adfm.201806482>
34. **Anurag Krishna**, Andrew C. Grimsdale, "Hole transporting materials for mesoscopic perovskite solar cells - towards a rational design?", **J. Mater. Chem. A**, 2017, 5 16446–16466. [Impact factor = 9.5] <https://doi.org/10.1039/C7TA01258F>
35. **Anurag Krishna**, Andrey V. Lunchev, Andrew C. Grimsdale "Suzuki Polycondensation", **Synthetic Methods for Conjugated Polymers and Carbon Materials.**, 2017, 59–95. <https://doi.org/10.1002/9783527695959.ch2>
36. Chao Shen, Marc Courté, **Anurag Krishna**, Shasha Tang, Denis Fichou, "Quinoidal 2,2',6,6'-Tetraphenyl-Dipyranylidene as a Dopant-Free Hole-Transport Material for Stable and Cost-Effective Perovskite Solar Cells", **Energy Technol.**, 2017, 5, (10) 1852-1858. [Impact factor = 3.6] <https://doi.org/10.1002/ente.201700448>
37. **Anurag Krishna**, Dharani Sabba, Jun Yin, Annalisa Bruno, Liisa J. Antila, Cesare Soci, Subodh Mhaisalkar, Andrew C. Grimsdale, "Facile synthesis of a hole transporting material with a silafluorene core for efficient mesoscopic CH₃NH₃PbI₃ perovskite solar cells", **J. Mater. Chem. A**, 2016, 4 8750–8754. [Impact factor = 9.1] <https://doi.org/10.1039/C6TA01776B>
38. **Anurag Krishna**, Dharani Sabba, Jun Yin, Annalisa Bruno, Pablo P. Boix, Yang Gao, Herlina A. Dewi, Gagik G. Gurzadyan, Cesare Soci, Subodh G. Mhaisalkar, Andrew C. Grimsdale, "Facile Synthesis of a Furan–Arylamine Hole-Transporting Material for High-Efficiency, Mesoscopic Perovskite Solar Cells", **Chem. Eur. J.**, 2015, 21 (43) 15113–15117. [Impact factor = 3.7] <https://doi.org/10.1002/chem.201503099>
39. Jun Yin, Daniele Cortecchia, **Anurag Krishna**, Shi Chen, Nripan Mathews, Andrew C. Grimsdale, Cesare Soci, "Interfacial Charge Transfer Anisotropy in Polycrystalline Lead Iodide Perovskite Films", **J. Phys. Chem. Lett.**, 2015, 6 (8) 1396–1402. [Impact factor = 4.6] <https://doi.org/10.1021/acs.jpclett.5b00431>
40. Teddy Salim, Shuangyong Sun, Yuichiro Abe, **Anurag Krishna**, Andrew C. Grimsdale, Yeng Ming Lam, "Perovskite-based solar cells: impact of morphology and device architecture on device performance", **J. Mater. Chem. A**, 2015, 3 8943–8969. [Impact factor = 9.1] <https://doi.org/10.1039/C4TA05226A>
41. **Anurag Krishna**, Dharani Sabba, Hairong Li, Jun Yin, Pablo P. Boix, Cesare Soci, Subodh G. Mhaisalkar, Andrew C. Grimsdale, "Novel hole transporting materials based on triptycene core for high efficiency mesoscopic perovskite solar cells", **Chem. Sci.**, 2014, 5 2702–2709. [Impact factor = 7.6] <https://doi.org/10.1039/C4SC00814F>
42. **Anurag Krishna**, Cosmin Laslau, Geoffrey Waterhouse, Zoran Zujovic, Jadranka Travas-Sejdic, "Effect of ionic liquid on polyaniline chemically synthesised under falling-pH conditions", **Chemical Papers**, 2013, 67 (8). [Impact factor = 2.5] <https://doi.org/10.2478/s11696-013-0327-7doi.org/10.1038/s44359-026-00173-2>



Scopus

KEY EUROPEAN & NATIONAL RESEARCH PROJECTS

EIC Transition: TRANSPIRE-Valorisation of imec's perovskite technology: bifacial perovskite and perovskite–silicon tandem modules from TRL 4 to pilot-line readiness. Grant Value: ~ €2.5 M. Duration: 2026-2029. Role: Coordinator and principal investigator (<https://cordis.europa.eu/project/id/101291137>)

Horizon Europe Innovation Action: LAPERITIVO-Large-Area Perovskite Solar Module Manufacturing with High Efficiency, Long-Term Stability and Low Environmental Impact. Grant Value: ~ €10 M. Duration: 01/09/2024-29/02/2028. Role: part of the coordinating team and principal investigator (<https://cordis.europa.eu/project/id/101147311>)

Horizon Europe CSA: TESTARE-Twinning for excellence in Testing new generation PV: Long-term STABILITY and field Reliability. Grant Value: ~ €1.5 M. Duration: 01/01/2023 to 30/06/2026. Role: One of the partners, and principal investigator (<https://cordis.europa.eu/project/id/101079488>)

Horizon Europe CSA: APPROACH-Advanced Photonic PRocesses for novel sOlar energy hArvesting teCHnologies. Grant Value: ~ € 2.98 M. Duration: 01/06/2023 to 30/11/2026. Role: One of the partners, and Principal investigator from imec. (<https://cordis.europa.eu/project/id/101120397>)

Marie Skłodowska-Curie Postdoctoral fellowship: MOLMAPS - MOlecular Materials for Passivation in large Area Perovskite Solar modules- Grant value: ~€191760. Duration: 09/2019 to 08/2021. Role: Supervisor (<https://cordis.europa.eu/project/id/101153019>)

Marie Skłodowska-Curie Postdoctoral fellowship: ARMOR. Role: Supervisor. (<https://cordis.europa.eu/project/id/101281112>)

Marie Skłodowska-Curie Postdoctoral fellowship: SHINE. Role: Supervisor. (<https://cordis.europa.eu/project/id/101281076>)

Marie Skłodowska-Curie individual fellowship: STARS - Stable perovskite solar cells via interfacial engineering of 2D/3D mixed-dimensional Absorbers and Robust dopant-free hole transporting materials. Grant value: ~€191149. Duration: 09/2019 to 08/2021. Role: Recipient and principal investigator (<https://cordis.europa.eu/project/id/843453>)

Research Foundation – Flanders (FWO): PROCEED - Hybrid perovskites as a material platform for the conversion, emission, and detection of light (PROCEED). Grant value: ~ € 2.38 M. Duration: 01/01/2020 to 31/12/2023. Role: One of the partners, and the principal investigator from imec.

Federal Energy Transition Fund: PROCURA-Power to X and Carbon Capture & Utilization Roadmap for Belgium. Grant value: ~€4.5 M. Duration: 01/03/2020 to 28/02/2025. Role: One of the partners.

WORK HIGHLIGHT IN MEDIA/PRESS

- [Yahoo Tech \(2025\)](#) – Global coverage of imec’s advances in perovskite PV reliability.
- [TechXplore \(2025\)](#) – Report on breakthrough outdoor stability of perovskite solar modules.
- [eeNews Europe \(2025\)](#) – imec demonstrates long-term outdoor stability of perovskite modules.
- [PV Magazine \(2025\)](#) – Field testing shows perovskite solar modules retain 78% efficiency after one year.
- [PV Magazine \(2023\)](#) – imec develops 781 cm² bifacial perovskite solar panel.
- [PV Magazine \(2023\)](#) – imec presents inverted perovskite solar cell with 24.3% efficiency.
- [R&D World \(2023\)](#) – imec progress in developing scalable perovskite solar modules.
- [De Tijd \(Belgium, 2025\)](#) – Belgian media feature on imec’s “super solar panels” boosting power output.
- [PV Magazine \(2024\)](#) – BECOME PV workshop on commercialization of emerging PV technologies.
- [PV Magazine \(2021\)](#) – Interface engineering enables stable 23.4%-efficient perovskite solar cell.

PROFESSIONAL SERVICE & ORGANISATION

- Academic Editor, *Scientific Reports* (Nature Portfolio)
- Member, Review College of the Research Foundation – Flanders (FWO); evaluator for Czech Academy of Sciences institute evaluations.
- Member, ETIP PV Working Group, contributing to European PV roadmaps and strategy documents.
- Co-organiser of international events including COMPER24 and RECIPE25 (MATSUS conferences), BECOME PV and FuturePV workshops, and the SENECA seminar series on solar energy conversion and storage.

TEACHING, SUPERVISION & MENTORING

- Supervision at imec: currently supervising 3 MSCA postdoctoral fellows, 2 PhD students, 2 postdocs, and several MSc interns; previously mentored 3 PhD students on perovskite and tandem module R&D.
- EPFL and CNRS: mentored and co-supervised PhD and master’s students in perovskite stability, interface engineering, and device physics.
- NTU Singapore: co-supervised 4 bachelor’s theses and 1 master’s project in organic electronics and solar cells.
- Teaching: graduate teaching assistant for “Environmental Sustainability & Materials” (NTU); guest lectures on advanced photovoltaics at UHasselt; completed a diploma course in university teaching and pedagogy.

REFERENCES

- Prof. Michael Graetzel, Director, Laboratory of Photonics and Interfaces (LPI), Ecole Polytechnique de Lausanne (EPFL), Switzerland. Email: michael.graetzel@epfl.ch
- Prof. Jef Poortmans, Scientific Director Energy and imec Fellow, Interuniversity Microelectronics Centre (imec), Belgium. Email: jef.poortmans@imec.be
- Prof. Anders Hagfeldt, Vice Chancellor, Uppsala University, Sweden. Email: anders.hagfeldt@uu.se